

RAIR Lab Publication List

2024

- [1] James T. Oswald, Thomas M. Ferguson, and Selmer Bringsjord. “A Universal Intelligence Measure for Arithmetical Uncomputable Environments”. In: *Artificial General Intelligence - 17th International Conference, AGI 2024, Seattle, WA, USA, August 13-16, 2024, Proceedings*. Ed. by Kristinn R. Thórisson, Peter Isaev, and Arash Sheikhlari. Vol. 14951. Lecture Notes in Computer Science. Springer, 2024, pp. 134–144. DOI: [10.1007/978-3-031-65572-2_15](https://doi.org/10.1007/978-3-031-65572-2_15). URL: https://doi.org/10.1007/978-3-031-65572-2_15.

2023

- [1] Vivek Nallur, Louise A. Dennis, Selmer Bringsjord, and Naveen Sundar Govindarajulu. “A Partially Synthesized Position on the Automation of Machine Ethics”. In: *Digit. Soc.* 2.2 (2023). DOI: [10.1007/s44206-023-00040-8](https://doi.org/10.1007/s44206-023-00040-8). URL: <https://doi.org/10.1007/s44206-023-00040-8>.
- [2] Selmer Bringsjord, Michael Giancola, Naveen Sundar Govindarajulu, John Slowik, James T. Oswald, Paul Bello, and Micah Clark. “Argument-based inductive logics, with coverage of compromised perception”. In: *Frontiers Artif. Intell.* 6 (2023). DOI: [10.3389/frai.2023.1144569](https://doi.org/10.3389/frai.2023.1144569). URL: <https://doi.org/10.3389/frai.2023.1144569>.
- [3] Selmer Bringsjord, John Slowik, Naveen Sundar Govindarajulu, Michael Giancola, James T. Oswald, and Rikhiya Ghosh. “Affect-based Planning for a Meta-Cognitive Robot Sculptor: First Steps”. In: *11th International Conference on Affective Computing and Intelligent Interaction, ACII 2023 - Workshops and Demos, Cambridge, MA, USA, September 10-13, 2023*. IEEE, 2023, pp. 1–8. DOI: [10.1109/ACIIW59127.2023.10388202](https://doi.org/10.1109/ACIIW59127.2023.10388202). URL: <https://doi.org/10.1109/ACIIW59127.2023.10388202>.
- [4] Selmer Bringsjord, James T. Oswald, Michael Giancola, Brandon Rozek, and Naveen Sundar Govindarajulu. “The M Cognitive Meta-architecture as Touchstone for Standard Modeling of AGI-Level Minds”. In: *Artificial General Intelligence - 16th International Conference, AGI 2023, Stockholm, Sweden, June 16-19, 2023, Proceedings*. Ed. by Patrick Hammer, Marjan Alirezaie, and Claes Strannegård. Vol. 13921. Lecture Notes in Computer Science. Springer, 2023, pp. 62–73. DOI: [10.1007/978-3-031-33469-6_7](https://doi.org/10.1007/978-3-031-33469-6_7). URL: https://doi.org/10.1007/978-3-031-33469-6_7.
- [5] James T. Oswald and Brandon Rozek. “Parallel Verification of Natural Deduction Proof Graphs”. In: *Proceedings of the 18th International Workshop on Logical Frameworks and Meta-Languages: Theory and Practice, LFMTP@FSCD 2023, Rome, Italy, 2nd July 2023*. Ed. by Alberto Ciaffaglione and Carlos Olarte. Vol. 396. EPTCS. 2023, pp. 36–51. DOI: [10.4204/EPTCS.396.4](https://doi.org/10.4204/EPTCS.396.4). URL: <https://doi.org/10.4204/EPTCS.396.4>.

RAIR Lab Publication List

2022

- [1] Selmer Bringsjord, Naveen Sundar Govindarajulu, John Slowik, James T. Oswald, Michael Giancola, John Angel, Shreya Banerjee, and Aidan Flaherty. “PERI.2 Goes to PreSchool and Beyond, in Search of AGI”. In: *Artificial General Intelligence - 15th International Conference, AGI 2022, Seattle, WA, USA, August 19-22, 2022, Proceedings*. Ed. by Ben Goertzel, Matt Iklé, Alexey Potapov, and Denis K. Ponomaryov. Vol. 13539. Lecture Notes in Computer Science. Springer, 2022, pp. 178–187. DOI: [10 . 1007 / 978 - 3 - 031 - 19907 - 3 _ 17](https://doi.org/10.1007/978-3-031-19907-3_17). URL: https://doi.org/10.1007/978-3-031-19907-3%5C_17.
- [2] Michael Giancola, Selmer Bringsjord, and Naveen Sundar Govindarajulu. “Toward Generating Natural-Language Explanations of Modal-Logic Proofs”. In: *Artificial General Intelligence - 15th International Conference, AGI 2022, Seattle, WA, USA, August 19-22, 2022, Proceedings*. Ed. by Ben Goertzel, Matt Iklé, Alexey Potapov, and Denis K. Ponomaryov. Vol. 13539. Lecture Notes in Computer Science. Springer, 2022, pp. 220–230. DOI: [10 . 1007 / 978 - 3 - 031 - 19907 - 3 _ 21](https://doi.org/10.1007/978-3-031-19907-3_21). URL: https://doi.org/10.1007/978-3-031-19907-3%5C_21.
- [3] Shreya Banerjee, Selmer Bringsjord, Michael Giancola, and Naveen Sundar Govindarajulu. “Qualitative Mechanical Problem-Solving by Artificial Agents: Further Progress, Under Psychometric AI”. In: *Proceedings of the Thirty-Fifth International Florida Artificial Intelligence Research Society Conference, FLAIRS 2022, Hutchinson Island, Jensen Beach, Florida, USA, May 15-18, 2022*. Ed. by Roman Barták, Fazel Keshtkar, and Michael Franklin. 2022. DOI: [10.32473/FLAIRS.V35I.130630](https://doi.org/10.32473/FLAIRS.V35I.130630). URL: <https://doi.org/10.32473/flairs.v35i.130630>.
- [4] Michael Giancola, Selmer Bringsjord, and Naveen Sundar Govindarajulu. “Novel Intensional Defeasible Reasoning for AI: Is it Cognitively Adequate? (poster)”. In: *Proceedings of the Workshop on Cognitive Aspects of Knowledge Representation co-located with the 31st international joint conference on artificial intelligence (IJCAI-ECAI 2022), Vienna, Austria, July 23, 2022*. Ed. by Jesse Heyninck, Thomas Meyer, Marco Ragni, Matthias Thimm, and Gabriele Kern-Isberner. Vol. 3251. CEUR Workshop Proceedings. CEUR-WS.org, 2022. URL: <https://ceur-ws.org/Vol-3251/paper9.pdf>.

2021

- [1] Selmer Bringsjord, Naveen Sundar Govindarajulu, and Michael Giancola. “Automated argument adjudication to solve ethical problems in multi-agent environments”. In: *Paladyn J. Behav. Robotics* 12.1 (2021), pp. 310–335. DOI: [10 . 1515 / PJBR - 2021 - 0009](https://doi.org/10.1515/PJBR-2021-0009). URL: <https://doi.org/10.1515/pjbr-2021-0009>.
- [2] Selmer Bringsjord and Naveen Sundar Govindarajulu. “Fundamental Proof Methods in Computer Science: A Computer-Based Approach, by Arkoudas and Musser, The MIT Press, Cambridge, USA, ISBN 978-0-262-03553-8”. In: *Theory Pract. Log. Program.* 21.2 (2021), pp. 283–290. DOI: [10 . 1017 / S1471068420000071](https://doi.org/10.1017/S1471068420000071). URL: <https://doi.org/10.1017/S1471068420000071>.
- [3] Selmer Bringsjord, Naveen Sundar Govindarajulu, and Michael Giancola. “AI Can Stop Mass Shootings, and More”. In: *CoRR* abs/2102.09343 (2021). arXiv: [2102.09343](https://arxiv.org/abs/2102.09343). URL: <https://arxiv.org/abs/2102.09343>.

RAIR Lab Publication List

2020

- [1] Selmer Bringsjord and Naveen Sundar Govindarajulu. “Rectifying the Mischaracterization of Logic by Mental Model Theorists”. In: *Cogn. Sci.* 44.12 (2020). DOI: [10.1111/COGS.12898](https://doi.org/10.1111/COGS.12898). URL: <https://doi.org/10.1111/cogs.12898>.
- [2] Selmer Bringsjord and Naveen Sundar G. “The Theory of Cognitive Consciousness, and Λ (Lambda)”. In: *J. Artif. Intell. Conscious.* 7.2 (2020), pp. 155–181. DOI: [10.1142/S2705078520500095](https://doi.org/10.1142/S2705078520500095). URL: <https://doi.org/10.1142/s2705078520500095>.
- [3] Selmer Bringsjord, Naveen Sundar Govindarajulu, John Licato, and Michael Giancola. “Learning Ex Nihilo”. In: *6th Global Conference on Artificial Intelligence, GCAI 2020, Hangzhou, China, April 6-9, 2020*. Ed. by Grégoire Danoy, Jun Pang, and Geoff Sutcliffe. Vol. 72. EPiC Series in Computing. EasyChair, 2020, pp. 1–27. DOI: [10.29007/GGCF](https://doi.org/10.29007/GGCF). URL: <https://doi.org/10.29007/ggcf>.
- [4] Michael Giancola, Selmer Bringsjord, Naveen Sundar Govindarajulu, and John Licato. “Adjudication of Symbolic & Connectionist Arguments in Autonomous Driving AI”. In: *6th Global Conference on Artificial Intelligence, GCAI 2020, Hangzhou, China, April 6-9, 2020*. Ed. by Grégoire Danoy, Jun Pang, and Geoff Sutcliffe. Vol. 72. EPiC Series in Computing. EasyChair, 2020, pp. 28–33. DOI: [10.29007/K647](https://doi.org/10.29007/K647). URL: <https://doi.org/10.29007/k647>.
- [5] Selmer Bringsjord, Michael Giancola, and Naveen Sundar Govindarajulu. “Culturally Aware Social Robots That Carry Humans Inside Them, Protected by Defeasible Argumentation Systems”. In: *Culturally Sustainable Social Robotics - Proceedings of Robophilosophy 2020, Virtual Event, 2020*. Ed. by Marco Nørskov, Johanna Seibt, and Oliver Santiago Quick. Vol. 335. Frontiers in Artificial Intelligence and Applications. IOS Press, 2020, pp. 440–456. DOI: [10.3233/FAIA200941](https://doi.org/10.3233/FAIA200941). URL: <https://doi.org/10.3233/FAIA200941>.

2019

- [1] Selmer Bringsjord and Naveen Sundar Govindarajulu. “Introducing Λ for Measuring Cognitive Consciousness”. In: *Papers of the 2019 Towards Conscious AI Systems Symposium co-located with the Association for the Advancement of Artificial Intelligence 2019 Spring Symposium Series (AAAI SSS-19), Stanford, CA, March 25-27, 2019*. Ed. by Antonio Chella, David Gamez, Patrick Lincoln, Riccardo Manzotti, and Jonathan D. Pfautz. Vol. 2287. CEUR Workshop Proceedings. CEUR-WS.org, 2019. URL: <https://ceur-ws.org/Vol-2287/paper26.pdf>.
- [2] Naveen Sundar Govindarajulu and Selmer Bringsjord. “Towards a Computable & Harnessable Model of Consciousness”. In: *Papers of the 2019 Towards Conscious AI Systems Symposium co-located with the Association for the Advancement of Artificial Intelligence 2019 Spring Symposium Series (AAAI SSS-19), Stanford, CA, March 25-27, 2019*. Ed. by Antonio Chella, David Gamez, Patrick Lincoln, Riccardo Manzotti, and Jonathan D. Pfautz. Vol. 2287. CEUR Workshop Proceedings. CEUR-WS.org, 2019. URL: <https://ceur-ws.org/Vol-2287/paper27.pdf>.

RAIR Lab Publication List

- [3] Naveen Sundar Govindarajulu, Jean-Claude Paquin, Shreya Banerjee, Atriya Sen, Paul Mayol, and Selmer Bringsjord. "On Datasets for Evaluating Architectures for Learning to Reason". In: *Proceedings of the AAAI 2019 Spring Symposium on Combining Machine Learning with Knowledge Engineering (AAAI-MAKE 2019) Stanford University, Palo Alto, California, USA, March 25-27, 2019., Stanford University, Palo Alto, California, USA, March 25-27, 2019*. Ed. by Andreas Martin, Knut Hinkelmann, Aurore Gerber, Doug Lenat, Frank van Harmelen, and Peter Clark. Vol. 2350. CEUR Workshop Proceedings. CEUR-WS.org, 2019. URL: <https://ceur-ws.org/Vol-2350/xposter4.pdf>.
- [4] Naveen Sundar Govindarajulu, Selmer Bringsjord, Rikhiya Ghosh, and Vasanth Sarathy. "Toward the Engineering of Virtuous Machines". In: *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society, AIES 2019, Honolulu, HI, USA, January 27-28, 2019*. Ed. by Vincent Conitzer, Gillian K. Hadfield, and Shannon Vallor. ACM, 2019, pp. 29–35. DOI: [10.1145/3306618.3314256](https://doi.org/10.1145/3306618.3314256). URL: <https://doi.org/10.1145/3306618.3314256>.
- [5] Selmer Bringsjord, Naveen Sundar Govindarajulu, and Christina Elmore. "Logicist Computational Cognitive Modeling of Infinitary False Belief Tasks". In: *Proceedings of the 41th Annual Meeting of the Cognitive Science Society, CogSci 2019: Creativity + Cognition + Computation, Montreal, Canada, July 24-27, 2019*. Ed. by Ashok K. Goel, Colleen M. Seifert, and Christian Freksa. cognitivesciencesociety.org, 2019, pp. 43–44. URL: <https://mindmodeling.org/cogsci2019/papers/0022/index.html>.
- [6] Selmer Bringsjord, Naveen Sundar Govindarajulu, Atriya Sen, and Christina Elmore. "A Formalization of Cognitive Continuity/Discontinuity, to Settle the Darwin's-Mistake Debate". In: *Proceedings of the 41th Annual Meeting of the Cognitive Science Society, CogSci 2019: Creativity + Cognition + Computation, Montreal, Canada, July 24-27, 2019*. Ed. by Ashok K. Goel, Colleen M. Seifert, and Christian Freksa. cognitivesciencesociety.org, 2019, p. 3417. URL: <https://mindmodeling.org/cogsci2019/papers/0725/index.html>.
- [7] Naveen Sundar Govindarajulu, Selmer Bringsjord, and Matthew Peveler. "On Quantified Modal Theorem Proving for Modeling Ethics". In: *Proceedings of the Second International Workshop on Automated Reasoning: Challenges, Applications, Directions, Exemplary Achievements, ARCADE@CADE 2019, Natal, Brazil, August 26, 2019*. Ed. by Martin Suda and Sarah Winkler. Vol. 311. EPTCS. 2019, pp. 43–49. DOI: [10.4204/EPTCS.311.7](https://doi.org/10.4204/EPTCS.311.7). URL: <https://doi.org/10.4204/EPTCS.311.7>.
- [8] Selmer Bringsjord and Naveen Sundar Govindarajulu. "Learning Ex Nihilo". In: *CoRR* abs/1903.03515 (2019). arXiv: [1903.03515](https://arxiv.org/abs/1903.03515). URL: <http://arxiv.org/abs/1903.03515>.

2018

- [1] Atriya Sen, Selmer Bringsjord, Naveen Sundar Govindarajulu, Paul Mayol, Rikhiya Ghosh, Biplav Srivastava, and Kartik Talamadupula. "Toward a Smart City Using Tentacular AI". In: *Ambient Intelligence - 14th European Conference, Aml 2018, Larnaca, Cyprus, November 12-14, 2018, Proceedings*. Ed. by Achilles Kameas and Kostas Stathis. Vol. 11249. Lecture Notes in Computer Science. Springer, 2018, pp. 106–112. DOI: [10.1007/978-3-030-03062-9_9](https://doi.org/10.1007/978-3-030-03062-9_9). URL: https://doi.org/10.1007/978-3-030-03062-9_9.

RAIR Lab Publication List

- [2] Selmer Bringsjord, Naveen Sundar G., Bertram F. Malle, and Matthias Scheutz. "Contextual Deontic Cognitive Event Calculi for Ethically Correct Robots". In: *International Symposium on Artificial Intelligence and Mathematics, ISAIM 2018, Fort Lauderdale, Florida, USA, January 3-5, 2018*. 2018. URL: https://isaim2018.cs.ou.edu/papers/ISAIM2018%5C_Ethics%5C_Bringsjord%5C_et al.pdf.
- [3] Naveen Sundar Govindarajulu, Rikhiya Ghosh, and Selmer Bringsjord. "Extending Formal Models of the Doctrine of Double Effect with Emotions". In: *International Symposium on Artificial Intelligence and Mathematics, ISAIM 2018, Fort Lauderdale, Florida, USA, January 3-5, 2018*. 2018. URL: https://isaim2018.cs.ou.edu/papers/ISAIM2018%5C_Ethics%5C_Govindarajulu%5C_et al.pdf.
- [4] John Angel, Naveen Sundar Govindarajulu, and Selmer Bringsjord. "Toward Formalizing Teleportation of Pedagogical Artificial Agents". In: *CoRR* abs/1804.03342 (2018). arXiv: 1804.03342. URL: <http://arxiv.org/abs/1804.03342>.
- [5] Naveen Sundar Govindarajulu, Selmer Bringsjord, and Rikhiya Ghosh. "One Formalization of Virtue Ethics via Learning". In: *CoRR* abs/1805.07797 (2018). arXiv: 1805.07797. URL: <http://arxiv.org/abs/1805.07797>.
- [6] Selmer Bringsjord, Naveen Sundar Govindarajulu, Atriya Sen, Matthew Peveler, Biplav Srivastava, and Kartik Talamadupula. "Tentacular Artificial Intelligence, and the Architecture Thereof, Introduced". In: *CoRR* abs/1810.07007 (2018). arXiv: 1810.07007. URL: <http://arxiv.org/abs/1810.07007>.
- [7] Naveen Sundar Govindarajulu, Selmer Bringsjord, and Rikhiya Ghosh. "Toward the Engineering of Virtuous Machines". In: *CoRR* abs/1812.03868 (2018). arXiv: 1812.03868. URL: <http://arxiv.org/abs/1812.03868>.

2017

- [1] Paul Bello and Selmer Bringsjord. "Two Problems Afflicting the Search for a Standard Model of the Mind". In: *2017 AAAI Fall Symposia, Arlington, Virginia, USA, November 9-11, 2017*. AAAI Press, 2017, pp. 296–301. URL: <https://aaai.org/ocs/index.php/FSS/FSS17/paper/view/15961>.
- [2] Naveen Sundar Govindarajulu and Selmer Bringsjord. "On Automating the Doctrine of Double Effect". In: *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence, IJCAI 2017, Melbourne, Australia, August 19-25, 2017*. Ed. by Carles Sierra. ijcai.org, 2017, pp. 4722–4730. DOI: 10.24963/IJCAI.2017/658. URL: <https://doi.org/10.24963/ijcai.2017/658>.
- [3] Selmer Bringsjord, Naveen Sundar Govindarajulu, Shreya Banerjee, and John Hummel. "Do Machine-Learning Machines Learn?" In: *Philosophy and Theory of Artificial Intelligence 2017, PT-AI 2017, Leeds, UK, November 4-5, 2017, Proceedings*. Ed. by Vincent C. Müller. Vol. 44. Studies in Applied Philosophy, Epistemology and Rational Ethics. Springer, 2017, pp. 136–157. DOI: 10.1007/978-3-319-96448-5_14. URL: https://doi.org/10.1007/978-3-319-96448-5%5C_14.
- [4] Naveen Sundar Govindarajulu and Selmer Bringsjord. "Proof Verification Can Be Hard!" In: *CoRR* abs/1703.08746 (2017). arXiv: 1703.08746. URL: <http://arxiv.org/abs/1703.08746>.

RAIR Lab Publication List

- [5] Naveen Sundar Govindarajulu and Selmer Bringsjord. “On Automating the Doctrine of Double Effect”. In: *CoRR* abs/1703.08922 (2017). arXiv: [1703.08922](https://arxiv.org/abs/1703.08922). URL: <http://arxiv.org/abs/1703.08922>.
- [6] Naveen Sundar Govindarajulu and Selmer Bringsjord. “Strength Factors: An Uncertainty System for a Quantified Modal Logic”. In: *CoRR* abs/1705.10726 (2017). arXiv: [1705.10726](https://arxiv.org/abs/1705.10726). URL: <http://arxiv.org/abs/1705.10726>.
- [7] Matthew Peveler, Biplav Srivastava, Kartik Talamadupula, Naveen Sundar G., Selmer Bringsjord, and Hui Su. “Towards Cognitive-and-Immersive Systems: Experiments in a Shared (or common) Blockworld Framework”. In: *CoRR* abs/1709.05958 (2017). arXiv: [1709.05958](https://arxiv.org/abs/1709.05958). URL: <http://arxiv.org/abs/1709.05958>.
- [8] Naveen Sundar Govindarajulu and Selmer Bringsjord. “Counterfactual Conditionals in Quantified Modal Logic”. In: *CoRR* abs/1710.04161 (2017). arXiv: [1710.04161](https://arxiv.org/abs/1710.04161). URL: <http://arxiv.org/abs/1710.04161>.

2016

- [1] Selmer Bringsjord and Atriya Sen. “On Creative Self-Driving Cars: Hire the Computational Logicians, Fast”. In: *Appl. Artif. Intell.* 30.8 (2016), pp. 758–786. DOI: [10.1080/08839514.2016.1229906](https://doi.org/10.1080/08839514.2016.1229906). URL: <https://doi.org/10.1080/08839514.2016.1229906>.
- [2] Nisar Ahmed, Paul Bello, Selmer Bringsjord, Micah Clark, Bradley Hayes, Christopher Miller, Frans A. Oliehoek, Frank Stein, and Matthijs T. J. Spaan. “The 2015 AAAI Fall Symposium Series Reports”. In: *AI Mag.* 37.2 (2016), pp. 85–90. DOI: [10.1609/aimag.v37i2.2661](https://doi.org/10.1609/aimag.v37i2.2661). URL: <https://doi.org/10.1609/aimag.v37i2.2661>.
- [3] Selmer Bringsjord, Rikhiya Ghosh, and James Pane-Joyce. “Deontic Counteridenticals and the Design of Ethically Correct Intelligent Agents: First Steps”. In: *Proceedings of the 1st Workshop on Ethics in the Design of Intelligent Agents, The Hague, The Netherlands, August 30, 2016*. Ed. by Grégory Bonnet, Maaïke Harbers, Koen V. Hindriks, Mike Katell, and Catherine Tessier. Vol. 1668. CEUR Workshop Proceedings. CEUR-WS.org, 2016, pp. 38–43. URL: <https://ceur-ws.org/Vol-1668/paper7.pdf>.
- [4] Naveen Sundar Govindarajulu and Selmer Bringsjord. “Crowdsourcing Theorem Proving via Natural Games”. In: *Proceedings of the Workshop on Bridging the Gap between Human and Automated Reasoning co-located with 25th International Joint Conference on Artificial Intelligence (IJCAI 2016), New York, USA, July 9, 2016*. Ed. by Claudia Schon and Ulrich Furbach. Vol. 1651. CEUR Workshop Proceedings. CEUR-WS.org, 2016, pp. 28–42. URL: <https://ceur-ws.org/Vol-1651/12340026.pdf>.

2015

- [1] Naveen Sundar Govindarajulu, Selmer Bringsjord, and Joshua Taylor. “Proof verification and proof discovery for relativity”. In: *Synth.* 192.7 (2015), pp. 2077–2094. DOI: [10.1007/s11229-014-0424-3](https://doi.org/10.1007/s11229-014-0424-3). URL: <https://doi.org/10.1007/s11229-014-0424-3>.

RAIR Lab Publication List

- [2] John Licato, Nick Marton, Boning Dong, Ron Sun, and Selmer Bringsjord. "Modeling the Creation and Development of Cause-Effect Pairs for Explanation Generation in a Cognitive Architecture". In: *Proceedings of the 3rd International Workshop on Artificial Intelligence and Cognition, Turin, Italy, September 28-29, 2015*. Ed. by Antonio Lieto, Cristina Battaglino, Daniele Paolo Radicioni, and Manuela Sanguinetti. Vol. 1510. CEUR Workshop Proceedings. CEUR-WS.org, 2015, pp. 29–39. URL: <https://ceur-ws.org/Vol-1510/paper3.pdf>.
- [3] Selmer Bringsjord, John Licato, Naveen Sundar Govindarajulu, Rikhiya Ghosh, and Atriya Sen. "Real robots that pass human tests of self-consciousness". In: *24th IEEE International Symposium on Robot and Human Interactive Communication, RO-MAN 2015, Kobe, Japan, August 31 - September 4, 2015*. IEEE, 2015, pp. 498–504. DOI: [10.1109/ROMAN.2015.7333698](https://doi.org/10.1109/ROMAN.2015.7333698). URL: <https://doi.org/10.1109/ROMAN.2015.7333698>.
- [4] Paul Bello, John Licato, and Selmer Bringsjord. "Constraints on freely chosen action for moral robots: Consciousness and control". In: *24th IEEE International Symposium on Robot and Human Interactive Communication, RO-MAN 2015, Kobe, Japan, August 31 - September 4, 2015*. IEEE, 2015, pp. 505–510. DOI: [10.1109/ROMAN.2015.7333654](https://doi.org/10.1109/ROMAN.2015.7333654). URL: <https://doi.org/10.1109/ROMAN.2015.7333654>.
- [5] Konner Atkin, John Licato, and Selmer Bringsjord. "Modeling interoperability between a reflex and reasoning system in a physical simulation environment". In: *Proceedings of the Poster Session and Student Colloquium Symposium, part of the 2015 Spring Simulation Multiconference, SpringSim '15, Alexandria, VA, USA, April 12-15, 2015*. Ed. by Salim Chemlal and Mohammed Moallemi. SCS/ACM, 2015, pp. 5–6. URL: <http://dl.acm.org/citation.cfm?id=2873016>.
- [6] Nick Marton, John Licato, and Selmer Bringsjord. "Creating and reasoning over scene descriptions in a physically realistic simulation". In: *Proceedings of the Symposium on Agent-Directed Simulation, part of the 2015 Spring Simulation Multiconference, SpringSim '15, Alexandria, VA, USA, April 12-15, 2015*. Ed. by Levent Yilmaz, Tuncer I. Ören, Gregory R. Madey, Maarten Sierhuis, and Yu Zhang. SCS/ACM, 2015, pp. 84–91. URL: <http://dl.acm.org/citation.cfm?id=2872549>.
- [7] Naveen Sundar Govindarajulu and Selmer Bringsjord. "Ethical Regulation of Robots Must Be Embedded in Their Operating Systems". In: *A Construction Manual for Robots' Ethical Systems - Requirements, Methods, Implementations*. Ed. by Robert Trappl. Cognitive Technologies. Springer, 2015, pp. 85–99. DOI: [10.1007/978-3-319-21548-8_5](https://doi.org/10.1007/978-3-319-21548-8_5). URL: https://doi.org/10.1007/978-3-319-21548-8_5.

2014

- [1] Selmer Bringsjord, Naveen Sundar Govindarajulu, Simon Ellis, Evan McCarty, and John Licato. "Nuclear deterrence and the logic of deliberative mindreading". In: *Cogn. Syst. Res.* 28 (2014), pp. 20–43. DOI: [10.1016/J.COGSYS.2013.08.001](https://doi.org/10.1016/j.cogsys.2013.08.001). URL: <https://doi.org/10.1016/j.cogsys.2013.08.001>.

RAIR Lab Publication List

- [2] Naveen Sundar Govindarajulu, John Licato, and Selmer Bringsjord. "Toward a Formalization of QA Problem Classes". In: *Artificial General Intelligence - 7th International Conference, AGI 2014, Quebec City, QC, Canada, August 1-4, 2014. Proceedings*. Ed. by Ben Goertzel, Laurent Orseau, and Javier Snider. Vol. 8598. Lecture Notes in Computer Science. Springer, 2014, pp. 228–233. DOI: [10.1007/978-3-319-09274-4_22](https://doi.org/10.1007/978-3-319-09274-4_22). URL: https://doi.org/10.1007/978-3-319-09274-4_22.
- [3] John Licato, Ron Sun, and Selmer Bringsjord. "Using Meta-Cognition for Regulating Explanatory Quality Through a Cognitive Architecture". In: *Proceedings of the Second International Workshop on Artificial Intelligence and Cognition (AIC 2014), Torino, Italy, November 26-27, 2014*. Ed. by Antonio Lieto, Daniele Paolo Radicioni, and Marco Cruciani. Vol. 1315. CEUR Workshop Proceedings. CEUR-WS.org, 2014, pp. 27–38. URL: <https://ceur-ws.org/Vol-1315/paper2.pdf>.
- [4] John Licato, Ron Sun, and Selmer Bringsjord. "Using a Hybrid Cognitive Architecture to Model Children's Errors in an Analogy Task". In: *Proceedings of the 36th Annual Meeting of the Cognitive Science Society, CogSci 2014, Quebec City, Canada, July 23-26, 2014*. Ed. by Paul Bello, Marcello Guarini, Marjorie McShane, and Brian Scassellati. cognitivesciencesociety.org, 2014. URL: <https://escholarship.org/uc/item/22b5d550>.
- [5] John Licato, Ron Sun, and Selmer Bringsjord. "Structural representation and reasoning in a hybrid cognitive architecture". In: *2014 International Joint Conference on Neural Networks, IJCNN 2014, Beijing, China, July 6-11, 2014*. IEEE, 2014, pp. 891–898. DOI: [10.1109/IJCNN.2014.6889895](https://doi.org/10.1109/IJCNN.2014.6889895). URL: <https://doi.org/10.1109/IJCNN.2014.6889895>.

2013

- [1] John Licato, Naveen Sundar Govindarajulu, Selmer Bringsjord, Michael Pomeranz, and Logan Gittelsohn. "Analogico-Deductive Generation of Gödel's First Incompleteness Theorem from the Liar Paradox". In: *IJCAI 2013, Proceedings of the 23rd International Joint Conference on Artificial Intelligence, Beijing, China, August 3-9, 2013*. Ed. by Francesca Rossi. IJCAI/AAAI, 2013, pp. 1004–1009. URL: <http://www.aaai.org/ocs/index.php/IJCAI/IJCAI13/paper/view/6988>.
- [2] Selmer Bringsjord and Naveen Sundar Govindarajulu. "Leibniz's Art of Infallibility, Watson, and the Philosophy, Theory, and Future of AI". In: *Fundamental Issues of Artificial Intelligence - 2nd Conference on Philosophy and Theory of Artificial Intelligence, PT-AI 2013, Oxford, UK, September 21-22, 2013, selected and invited papers*. Ed. by Vincent C. Müller. Vol. 376. Synthese Library. Springer, 2013, pp. 185–202. DOI: [10.1007/978-3-319-26485-1_12](https://doi.org/10.1007/978-3-319-26485-1_12). URL: https://doi.org/10.1007/978-3-319-26485-1_12.
- [3] Naveen Sundar Govindarajulu, John Licato, and Selmer Bringsjord. "Small Steps toward Hypercomputation via Infinitary Machine Proof Verification and Proof Generation". In: *Unconventional Computation and Natural Computation - 12th International Conference, UCNC 2013, Milan, Italy, July 1-5, 2013. Proceedings*. Ed. by Giancarlo Mauri, Alberto Dennunzio, Luca Manzoni, and Antonio E. Porreca. Vol. 7956. Lecture Notes in Computer Science. Springer, 2013, pp. 102–112. DOI: [10.1007/978-3-642-39074-6_11](https://doi.org/10.1007/978-3-642-39074-6_11). URL: https://doi.org/10.1007/978-3-642-39074-6_11.

RAIR Lab Publication List

2012

- [1] Selmer Bringsjord, Naveen Sundar G., Eugene Eberbach, and Yingrui Yang. "Perhaps the Rigorous Modeling of Economic Phenomena Requires Hypercomputation". In: *Int. J. Unconv. Comput.* 8.1 (2012), pp. 3–32. URL: <http://www.oldcitypublishing.com/journals/ijuc-home/ijuc-issue-contents/ijuc-volume-8-number-1-2012/ijuc-8-1-p-3-32/>.
- [2] Naveen Sundar Govindarajulu and Selmer Bringsjord. "The Myth of 'the Myth of Hypercomputation'". In: *Parallel Process. Lett.* 22.3 (2012). DOI: [10.1142/S0129626412400129](https://doi.org/10.1142/S0129626412400129). URL: <https://doi.org/10.1142/S0129626412400129>.
- [3] Selmer Bringsjord and Micah Henry Clark. "Red-Pill Robots Only, Please". In: *IEEE Trans. Affect. Comput.* 3.4 (2012), pp. 394–397. DOI: [10.1109/T-AFFC.2011.35](https://doi.org/10.1109/T-AFFC.2011.35). URL: <https://doi.org/10.1109/T-AFFC.2011.35>.

2011

- [1] Nate Chapin, Boleslaw K. Szymanski, Selmer Bringsjord, and Bettina Schimanski. "A bottom-up complement to the logic-based top-down approach to the story arrangement test". In: *J. Exp. Theor. Artif. Intell.* 23.3 (2011), pp. 329–341. DOI: [10.1080/0952813X.2010.502313](https://doi.org/10.1080/0952813X.2010.502313). URL: <https://doi.org/10.1080/0952813X.2010.502313>.
- [2] John Licato and Selmer Bringsjord. "In Defense of the Neo-Piagetian Approach to Modeling and Engineering Human-Level Cognitive Systems". In: *Advances in Cognitive Systems, Papers from the 2011 AAAI Fall Symposium, Arlington, Virginia, USA, November 4-6, 2011*. Vol. FS-11-01. AAAI Technical Report. AAAI, 2011. URL: <http://www.aaai.org/ocs/index.php/FSS/FSS11/paper/view/4202>.
- [3] Selmer Bringsjord, Colin Kuebler, Joshua Taylor, Griffin Milsap, Sean Austin, Jonas Braasch, Pauline Oliveros, Doug Van Nort, Adam Rosenkrantz, and Kasia Hayden. "Creativity and conducting: handle in the CAIRA project". In: *Proceedings of the 8th Conference on Creativity & Cognition, The High Museum of Art, Atlanta, Georgia, USA, November 3-6, 2011*. Ed. by Ashok K. Goel, D. Fox Harrell, Brian Magerko, Yukari Nagai, and Jane Prophet. ACM, 2011, pp. 319–320. DOI: [10.1145/2069618.2069676](https://doi.org/10.1145/2069618.2069676). URL: <https://doi.org/10.1145/2069618.2069676>.
- [4] Selmer Bringsjord and Naveen Sundar Govindarajulu. "Toward a Modern Geography of Minds, Machines, and Math". In: *Philosophy and Theory of Artificial Intelligence, PT-AI 2011, Thessaloniki, Greece, October 3-4, 2011, Proceedings*. Ed. by Vincent C. Müller. Vol. 5. Studies in Applied Philosophy, Epistemology and Rational Ethics. Springer, 2011, pp. 151–165. DOI: [10.1007/978-3-642-31674-6_11](https://doi.org/10.1007/978-3-642-31674-6_11). URL: https://doi.org/10.1007/978-3-642-31674-6_11.

2010

- [1] Selmer Bringsjord and Naveen Sundar G. "In Defense of the Unprovability of the Church-Turing Thesis". In: *Int. J. Unconv. Comput.* 6.5 (2010), pp. 353–373. URL: <http://www.oldcitypublishing.com/journals/ijuc-home/ijuc-issue-contents/ijuc-volume-6-number-5-2010/ijuc-6-5-p-353-373/>.

RAIR Lab Publication List

2009

- [1] Konstantine Arkoudas and Selmer Bringsjord. “Vivid: A framework for heterogeneous problem solving”. In: *Artif. Intell.* 173.15 (2009), pp. 1367–1405. DOI: [10.1016/J.ARTINT.2009.06.002](https://doi.org/10.1016/J.ARTINT.2009.06.002). URL: <https://doi.org/10.1016/j.artint.2009.06.002>.
- [2] Konstantine Arkoudas and Selmer Bringsjord. “Propositional Attitudes and Causation”. In: *Int. J. Softw. Informatics* 3.1 (2009), pp. 47–65. URL: http://www.ijsi.org/ch/reader/view%5C_abstract.aspx?file%5C_no=32%5C%5C#38;flag=1.
- [3] Konstantine Arkoudas, Selmer Bringsjord, and Sangeet Khemlani. “Qualitative Spatial Reasoning Via 3-Valued Heterogeneous Logic”. In: *KEOD 2009 - Proceedings of the International Conference on Knowledge Engineering and Ontology Development, Funchal - Madeira, Portugal, October 6-8, 2009*. Ed. by Jan L. G. Dietz. INSTICC Press, 2009, pp. 80–87.
- [4] Jacob Beal, Paul Bello, Nicholas L. Cassimatis, Michael H. Coen, Paul R. Cohen, Alex Davis, Mark T. Maybury, Alexei V. Samsonovich, Andrew Shilliday, Marjorie Skubic, Joshua Taylor, Sharon Walter, Patrick Henry Winston, and Beverly Park Woolf. “Reports of the AAAI 2008 Fall Symposia”. In: *AI Mag.* 30.2 (2009), pp. 106–111. DOI: [10.1609/aimag.v30i2.2231](https://doi.org/10.1609/aimag.v30i2.2231). URL: <https://doi.org/10.1609/aimag.v30i2.2231>.

2008

- [1] Selmer Bringsjord and Andrew Shilliday. “Organizing Committee”. In: *Automated Scientific Discovery, Papers from the 2008 AAAI Fall Symposium, Arlington, Virginia, USA, November 7-9, 2008*. Vol. FS-08-03. AAAI Technical Report. AAAI, 2008. URL: <http://www.aaai.org/Library/Symposia/Fall/2008/fs08-03-010.php>.
- [2] Joshua Taylor and Selmer Bringsjord. “Discovery Using Heterogeneous Combined Logics”. In: *Automated Scientific Discovery, Papers from the 2008 AAAI Fall Symposium, Arlington, Virginia, USA, November 7-9, 2008*. Vol. FS-08-03. AAAI Technical Report. AAAI, 2008, p. 30. URL: <http://www.aaai.org/Library/Symposia/Fall/2008/fs08-03-009.php>.
- [3] Selmer Bringsjord, Andrew Shilliday, Joshua Taylor, Dan Werner, Micah Clark, Ed Charpentier, and Alexander Bringsjord. “Toward Logic-Based Cognitively Robust Synthetic Characters in Digital Environments”. In: *Artificial General Intelligence 2008, Proceedings of the First AGI Conference, AGI 2008, March 1-3, 2008, University of Memphis, Memphis, TN, USA*. Ed. by Pei Wang, Ben Goertzel, and Stan Franklin. Vol. 171. Frontiers in Artificial Intelligence and Applications. IOS Press, 2008, pp. 87–98. URL: <http://www.booksonline.iospress.nl/Content/View.aspx?piid=8299>.
- [4] Konstantine Arkoudas and Selmer Bringsjord. “Toward Formalizing Common-Sense Psychology: An Analysis of the False-Belief Task”. In: *PRICAI 2008: Trends in Artificial Intelligence, 10th Pacific Rim International Conference on Artificial Intelligence, Hanoi, Vietnam, December 15-19, 2008. Proceedings*. Ed. by Tu Bao Ho and Zhi-Hua Zhou. Vol. 5351. Lecture Notes in Computer Science. Springer, 2008, pp. 17–29. DOI: [10.1007/978-3-540-89197-0_6](https://doi.org/10.1007/978-3-540-89197-0_6). URL: https://doi.org/10.1007/978-3-540-89197-0_6.

RAIR Lab Publication List

2007

- [1] Konstantine Arkoudas and Selmer Bringsjord. “Computers, Justification, and Mathematical Knowledge”. In: *Minds Mach.* 17.2 (2007), pp. 185–202. DOI: [10.1007/S11023-007-9063-5](https://doi.org/10.1007/S11023-007-9063-5). URL: <https://doi.org/10.1007/s11023-007-9063-5>.
- [2] Selmer Bringsjord, Konstantine Arkoudas, Micah Clark, Andrew Shilliday, Joshua Taylor, Bettina Schimanski, and Yingrui Yang. “Reporting on Some Logic-Based Machine Reading Research”. In: *Machine Reading, Papers from the 2007 AAAI Spring Symposium, Technical Report SS-07-06, Stanford, California, USA, March 26-28, 2007*. AAAI, 2007, pp. 23–28. URL: <http://www.aaai.org/Library/Symposia/Spring/2007/ss07-06-005.php>.
- [3] Joshua Taylor, Andrew Shilliday, and Selmer Bringsjord. “Provability-Based Semantic Interoperability Via Translation Graphs”. In: *Advances in Conceptual Modeling - Foundations and Applications, ER 2007 Workshops CMLSA, FP-UML, ONISW, QoIS, RIGiM, SeCoGIS, Auckland, New Zealand, November 5-9, 2007, Proceedings*. Ed. by Jean-Luc Hainaut, Elke A. Rundensteiner, Markus Kirchberg, Michela Bertolotto, Mathias Brochhausen, Yi-Ping Phoebe Chen, Samira Si-Said Cherfi, Martin Doerr, Hyoil Han, Sven Hartmann, Jeffrey Parsons, Geert Poels, Colette Rolland, Juan Trujillo, Eric S. K. Yu, and Esteban Zimányi. Vol. 4802. Lecture Notes in Computer Science. Springer, 2007, pp. 180–189. DOI: [10.1007/978-3-540-76292-8_21](https://doi.org/10.1007/978-3-540-76292-8_21). URL: https://doi.org/10.1007/978-3-540-76292-8_21.
- [4] Selmer Bringsjord, Konstantine Arkoudas, Deepa Mukherjee, Andrew Edward Shilliday, Joshua Taylor, Micah Henry Clark, and Elizabeth Bringsjord. “The Multi-Mind Effect”. In: *Proceedings of the 2007 International Conference on Artificial Intelligence, ICAI 2007, Volume I, June 25-28, 2007, Las Vegas, Nevada, USA*. Ed. by Hamid R. Arabnia, Mary Qu Yang, and Jack Y. Yang. CSREA Press, 2007, pp. 43–49.
- [5] Andrew Shilliday, Joshua Taylor, Micah Clark, and Selmer Bringsjord. “Provability-Based Semantic Interoperability for Information Sharing and Joint Reasoning”. In: *Ontologies and Semantic Technologies for Intelligence [papers from the Second International Ontology for the Intelligence Community (OIC) Conference, November 28-29, 2007, Columbia, MD, USA]*. Ed. by Leo Obrst, Terry L. Janssen, and Werner Ceusters. Vol. 213. Frontiers in Artificial Intelligence and Applications. IOS Press, 2007, pp. 109–128. DOI: [10.3233/978-1-60750-581-5-109](https://doi.org/10.3233/978-1-60750-581-5-109). URL: <https://doi.org/10.3233/978-1-60750-581-5-109>.

2006

- [1] Selmer Bringsjord, Owen Kellett, Andrew Shilliday, Joshua Taylor, Bram van Heuveln, Yingrui Yang, Jeffrey Baumes, and Kyle Ross. “A new Gödelian argument for hypercomputing minds based on the busy beaver problem”. In: *Appl. Math. Comput.* 176.2 (2006), pp. 516–530. DOI: [10.1016/J.AMC.2005.09.071](https://doi.org/10.1016/J.AMC.2005.09.071). URL: <https://doi.org/10.1016/j.amc.2005.09.071>.
- [2] Selmer Bringsjord, Konstantine Arkoudas, and Paul Bello. “Toward a General Logician Methodology for Engineering Ethically Correct Robots”. In: *IEEE Intell. Syst.* 21.4 (2006), pp. 38–44. DOI: [10.1109/MIS.2006.82](https://doi.org/10.1109/MIS.2006.82). URL: <https://doi.org/10.1109/MIS.2006.82>.
- [3] Yingrui Yang, Selmer Bringsjord, and Paul Bello. “The mental possible worlds mechanism and the lobster problem: an analysis of a complex GRE logical reasoning task”. In: *J. Exp. Theor. Artif. Intell.* 18.2 (2006), pp. 157–168. DOI: [10.1080/09528130600558125](https://doi.org/10.1080/09528130600558125). URL: <https://doi.org/10.1080/09528130600558125>.

RAIR Lab Publication List

- [4] Selmer Bringsjord and Micah Clark. “For Problems Sufficiently Hard ... AI Needs CogSci”. In: *Between a Rock and a Hard Place: Cognitive Science Principles Meet AI-Hard Problems, Papers from the 2006 AAAI Spring Symposium, Technical Report SS-06-02, Stanford, California, USA, March 27-29, 2006*. AAAI, 2006, pp. 23–26. URL: <http://www.aaai.org/Library/Symposia/Spring/2006/ss06-02-007.php>.

2004

- [1] Selmer Bringsjord and Konstantine Arkoudas. “The modal argument for hypercomputing minds”. In: *Theor. Comput. Sci.* 317.1-3 (2004), pp. 167–190. DOI: [10.1016/J.TCS.2003.12.010](https://doi.org/10.1016/J.TCS.2003.12.010). URL: <https://doi.org/10.1016/j.tcs.2003.12.010>.
- [2] Selmer Bringsjord and Bettina Schimanski. ““Pulling it All Together” via Psychometric AI”. In: *Achieving Human-Level Intelligence through Integrated Systems and Research, Papers from the 2004 AAAI Fall Symposium. Arlington, VA, USA, October 22-24, 2004*. Vol. FS-04-01. AAAI Press, 2004, pp. 9–16. URL: <https://www.aaai.org/Library/Symposia/Fall/2004/fs04-01-002.php>.
- [3] Konstantine Arkoudas and Selmer Bringsjord. “Metareasoning for Multi-agent Epistemic Logics”. In: *Computational Logic in Multi-Agent Systems, 5th International Workshop, CLIMA V, Lisbon, Portugal, September 29-30, 2004, Revised Selected and Invited Papers*. Ed. by João Alexandre Leite and Paolo Torroni. Vol. 3487. Lecture Notes in Computer Science. Springer, 2004, pp. 111–125. DOI: [10.1007/11533092_7](https://doi.org/10.1007/11533092_7). URL: https://doi.org/10.1007/11533092_5C_7.
- [4] Selmer Bringsjord and Joshua Taylor. “P=NP”. In: *CoRR* cs.CC/0406056 (2004). URL: <http://arxiv.org/abs/cs/0406056>.

2003

- [1] Paul Bello and Selmer Bringsjord. “HILBERT & PATRIC: Hybrid Intelligent Agent Technology for Teaching Context-Independent Reasoning”. In: *J. Educ. Technol. Soc.* 6.3 (2003), pp. 30–42. URL: https://www.j-ets.net/ETS/journals/6%5C_3/5.pdf.
- [2] Selmer Bringsjord and Bettina Schimanski. “What is Artificial Intelligence? Psychometric AI as an Answer”. In: *IJCAI-03, Proceedings of the Eighteenth International Joint Conference on Artificial Intelligence, Acapulco, Mexico, August 9-15, 2003*. Ed. by Georg Gottlob and Toby Walsh. Morgan Kaufmann, 2003, pp. 887–893. URL: <http://ijcai.org/Proceedings/03/Papers/128.pdf>.

2001

- [1] Selmer Bringsjord, Paul Bello, and David A. Ferrucci. “Creativity, the Turing Test, and the (Better) Lovelace Test”. In: *Minds Mach.* 11.1 (2001), pp. 3–27. DOI: [10.1023/A:1011206622741](https://doi.org/10.1023/A:1011206622741). URL: <https://doi.org/10.1023/A:1011206622741>.